

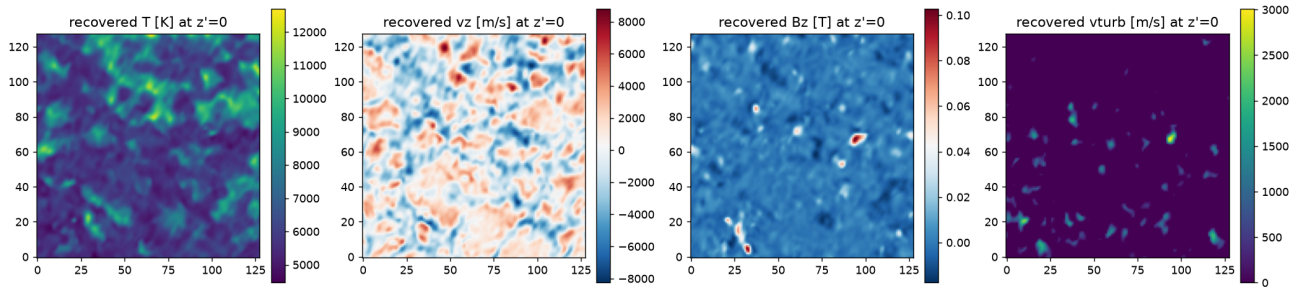
Adora PINN — Step 7 debug: the stripe artifact

As-built notes · Ivan Milic · 2026-09-20 · debugging the inversion

The step-7 inversion fits Stokes I well but the recovered maps show stripes. Investigating the saved field [/dat/milic/adora.pinn_develop/inverted_field.npz](#) and small controlled tests (the step-7 notebook itself was left untouched).

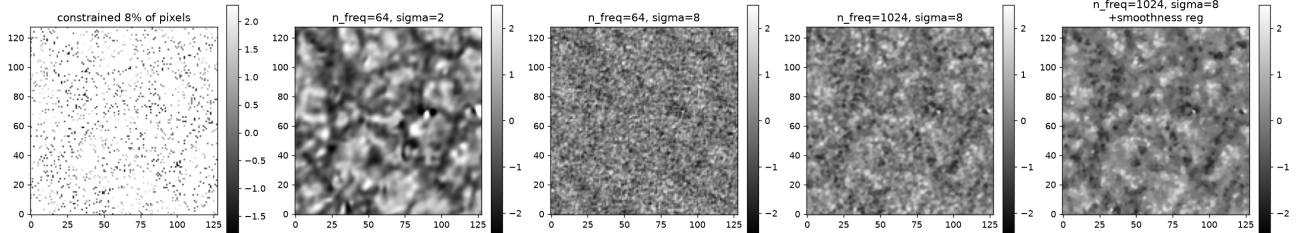
1. The artifact

The saved field has `n_freq=64`, $\sigma \approx (2.2, 2.0, 5.2)$ (the original bandwidth — the later (8,8,8) default has not been re-pretrained in). Its recovered maps at $z'=0$: T tracks the granulation; v_z **shows diagonal banding**; B_z is smooth with a few real concentrations but a faint banded background; v_{turb} **is sparse junk** (essentially unconstrained). A 2-D FFT of the maps shows all power — and the encoding's frequencies B_{xy} — confined to $|k| \lesssim 8 \text{ cyc/FOV}$, i.e. the field *cannot* represent finer horizontal structure.



2. Root cause (controlled test)

The stripes are the random Fourier modes **showing through wherever the data under-constrains the field** (weak channels, and the stochastic per-step pixel sampling). Fitting a fresh 2-D Fourier-MLP to only **8% of the observed-continuum pixels** (the inversion's sparse regime) makes the mechanism explicit:



- $\sigma=2$: smooth interpolation but blurry (few low-frequency modes $\rightarrow$ the gentle large-scale banding seen in v_z).
- $\sigma=8$: **fine high-frequency ringing** between the constrained points — *worse*. So simply raising σ to 8 would amplify the stripes.
- $\sigma=8$, `n_freq=1024`: still rings — **more features alone does not fix it**.
- $\sigma=8$ + a **gradient (smoothness) penalty**: clean *and* high-resolution — the cure.

(A full-batch fit, where every pixel is constrained, hides this: there $\sigma=2$ merely under-resolves, $\sigma=8$ is sharp. The stripes are purely an under-constrained-region effect.)

3. Recommendations

- **Add a spatial-smoothness regularizer** (gradient/TV penalty on the maps, at random points) — the most effective fix; it lets you use a high σ for resolution without ringing.
- **Match σ to the constraint density**: raising it to (8,8,8) *without* smoothness regularization will make the stripes worse, not better.
- **Denser fit sampling** (larger `m_fit`, or occasional full-map steps) — fewer under-constrained pixels per step.
- **Weakly-constrained channels** (v_{turb} , transverse B) should be regularized hard or frozen — they are currently free to absorb stripe junk.

Next concrete step (when you want it): add a smoothness term to `losses.py` and expose its weight in `run_inversion`; keep σ moderate and re-pretrain the warm start at the chosen bandwidth.